ECE343 Spring 2025

Objective 8.3: Maxwell's Equations



UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE

Lesson	Topics	Reading	Notes
24	BVP (Capacitor, Coax), Electric Potential	5.16	LO 5.3
25	Electric Boundary Conditions	5.18-5.22	LO 5.4
26	BC's/Capacitance	5.23-5.24	
27	Electrostatic Lab	Lab packet	
Chapter 6: Steady Current & Conductivity			
28	Current density, Conductivity, Resistance	Ch. 6	
Chapter 7: Magnetostatics			
29	Magnetic Fields, Gauss' Law for Magnetism	7.1-7.3	LO 7.1-7.2 **GR 2**
30	Ampere's Law	7.4-7.5, 7.9	LO 7.3
31	Ampere's Law Applied	7.6-7.8	LO 7.3
32	Magnetic Boundary Conditions	7.10-7.14	LO 7.4
33	Magnetostatics Lab	Lab packet	
Chapter 8: Time-Varying Fields			
34	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.1
35	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.2-8.3
Chapter 9: Plane Waves in Lossless Media			
36	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
37	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
38	Uniform Plane Waves & Polarization	9.4-9.6	LO 9.2-9.3
39	Normal Incidence	Vol 2, 5.6-5.7	LO 9.4-9.5
40	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
41	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
42	Incidence Lab	TBD	LO 9.5

Ch 8 Learning Objectives

7.1 I can articulate the significance of the displacement current and calculate it, given a time-varying electric field.

7.2 I can articulate the meaning of and calculate the relaxation time for a variety of materials.

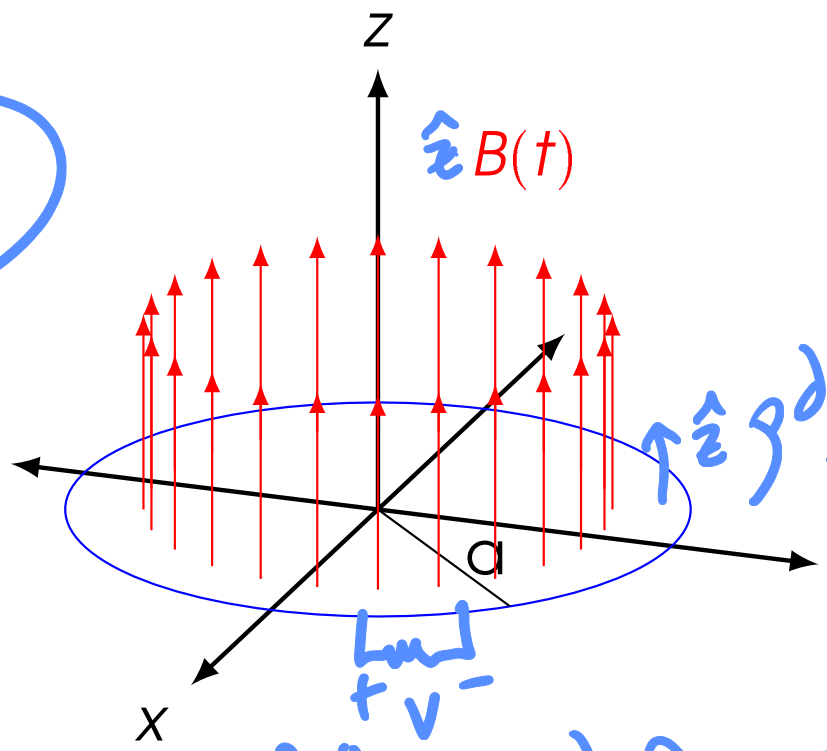
7.3 I know the forms of Maxwell's equations for time-varying fields and use them to calculate electric and magnetic fields for a variety of physical systems.

Maxwell's Equations

Integral Form	Point Form
$\iint \epsilon \vec{E} \cdot d\vec{S} = Q_{enc}$	$\nabla \cdot \vec{E} = \frac{\rho_v}{\epsilon}$
$\oint \vec{H} \cdot d\vec{S} = 0$	$\nabla \cdot \vec{H} = 0$
$\oint_C \vec{E} \cdot d\ell = -\frac{\partial \Psi_m}{\partial t}$ $\oint_C \vec{E} \cdot d\ell = -\iint \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$	$\nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}$
$\oint_C \vec{H} \cdot d\ell = I_{encl}$ Amp $\oint_C \vec{H} \cdot d\ell = \int \vec{J} \cdot d\vec{S} + \epsilon \frac{\partial}{\partial t} \iint \vec{E} \cdot d\vec{S}$	$\nabla \times \vec{H} = \sigma \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t}$

Maxwell's Equations: Example #1

A wire loop of radius $a = 10\text{cm}$ lies in the $x-y$ plane (i.e., $z = 0$) and contains a small gap with 1kΩ resistor. A time-varying magnetic flux density, $B(t)$ exists, with $t_0 = 3\text{s}$ and $B_0 = 0.2\text{T}$. Determine the potential and current across the resistor.



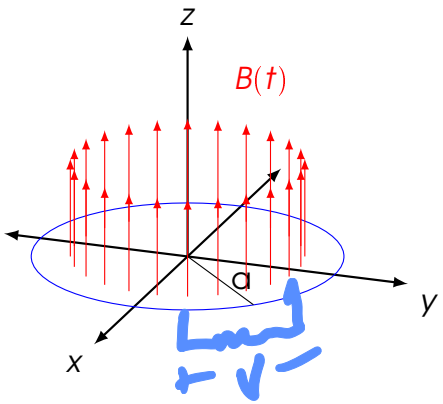
$$B(t) = \begin{cases} 0 & t < 0 \\ \dfrac{B_0}{t_0} t & 0 \leqslant t \leqslant t_0 \\ B_0 & t > t_0 \end{cases}$$

$$V = -\frac{\partial \psi_m}{\partial t} = -\frac{\partial}{\partial t}\int\int \vec{B}\cdot d\vec{s}$$

$$= -\int\int \frac{\partial}{\partial t}(\vec{B}\cdot d\vec{s})$$

$$= -\int\int \frac{\partial}{\partial t}\left(\hat{z}\frac{B_0 t}{t_0}\cdot \hat{z}\rho\, d\rho\, d\phi\right)$$

Maxwell's Equations: Example #1



$$V = -\iint \frac{\partial}{\partial t}\frac{B_0 t}{t_0}\rho\, d\rho\, d\phi$$

$$= -\iint \frac{B_0}{t_0}\rho\, d\rho\, d\phi$$

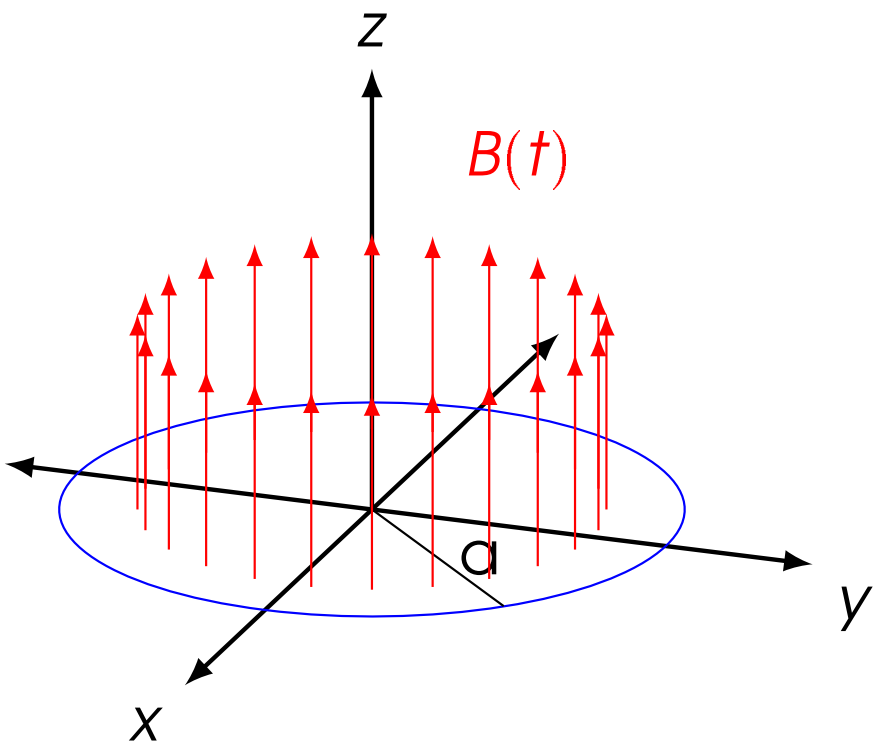
$$= -\frac{B_0}{t_0}\int_0^{2\pi}\int_0^{a}\rho\, d\rho\, d\phi$$

$$V = -\frac{B_0}{t_0}\pi a^2 = -\frac{(0.2\ \mathrm{T})}{2\mathrm{s}}\pi(0.1\mathrm{m})^2 = -2.1\mathrm{mV}$$

$$I = \frac{V}{R} = \frac{-2.1\mathrm{mV}}{1\mathrm{k}\Omega} = -2.1\mu\mathrm{A}$$

Maxwell's Equations: Example #2

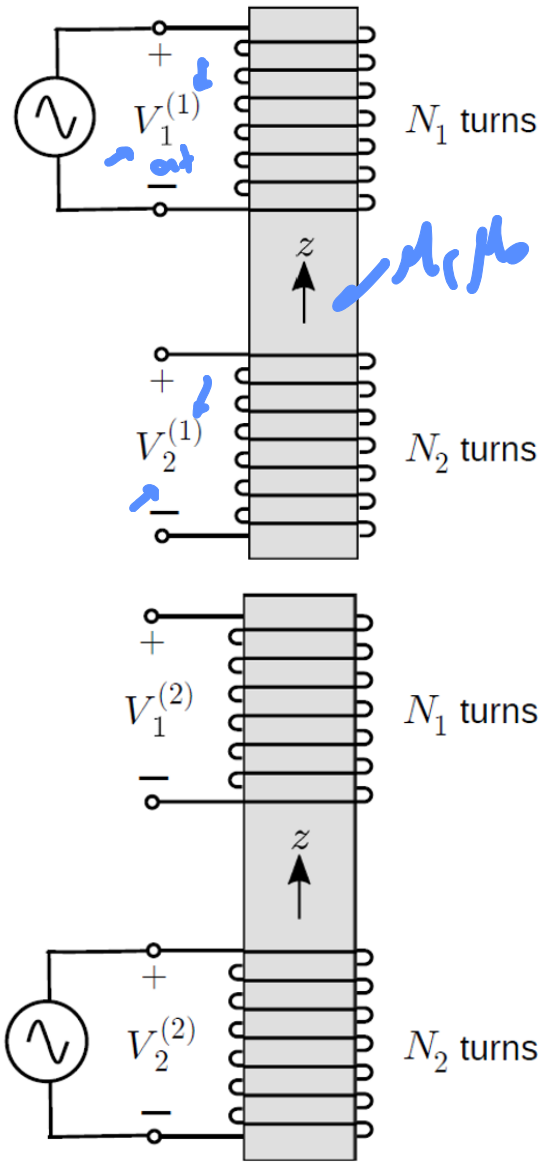
A wire loop of radius $a = 10$cm lies in the $x - y$ plane (i.e., $z = 0$) and contains a small gap with 1kΩ resistor. A time-varying magnetic flux density, $B(t)$ exists, with $t_0 = 3$s and $B_0 = 0.2$T. Determine the potential and current across the resistor.



$$B(t) = \sin(2\pi f t) \qquad f_0 = 1\text{kHz}$$

Maxwell's Equations: Example #2

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$$V_2' = -N_2 \frac{d\Psi_m}{dt} = -N_2 \frac{d}{dt}\iint \vec{B}\cdot d\vec{s}$$

$$d\vec{s} = -\hat{z}\,dA$$

$$V_2' = -N_2 \frac{d}{dt}\iint \vec{B}\cdot(-\hat{z}\,dA)$$

$$V_1^2 = -N_1 \frac{d\Psi_m}{dt} = -N_1 \frac{d}{dt}\iint \vec{B}\cdot d\vec{s}$$

$$d\vec{s} = \hat{z}\,dA$$

$$V_1^2 = -N_1 \frac{d}{dt}\iint \vec{B}\cdot(\hat{z}\,dA) = N_1 \frac{d}{dt}\iint \vec{B}\cdot(-\hat{z}\,dA)$$

$$\ddot{v} = -\frac{V_2'}{N_2} = \frac{V_1^2}{N_1} \Rightarrow \frac{V_2}{V_1} = \frac{N_2}{N_1} P \rightarrow \pm 1$$

$\frac{N_1}{N_2}\ddot{v}$ winding direction

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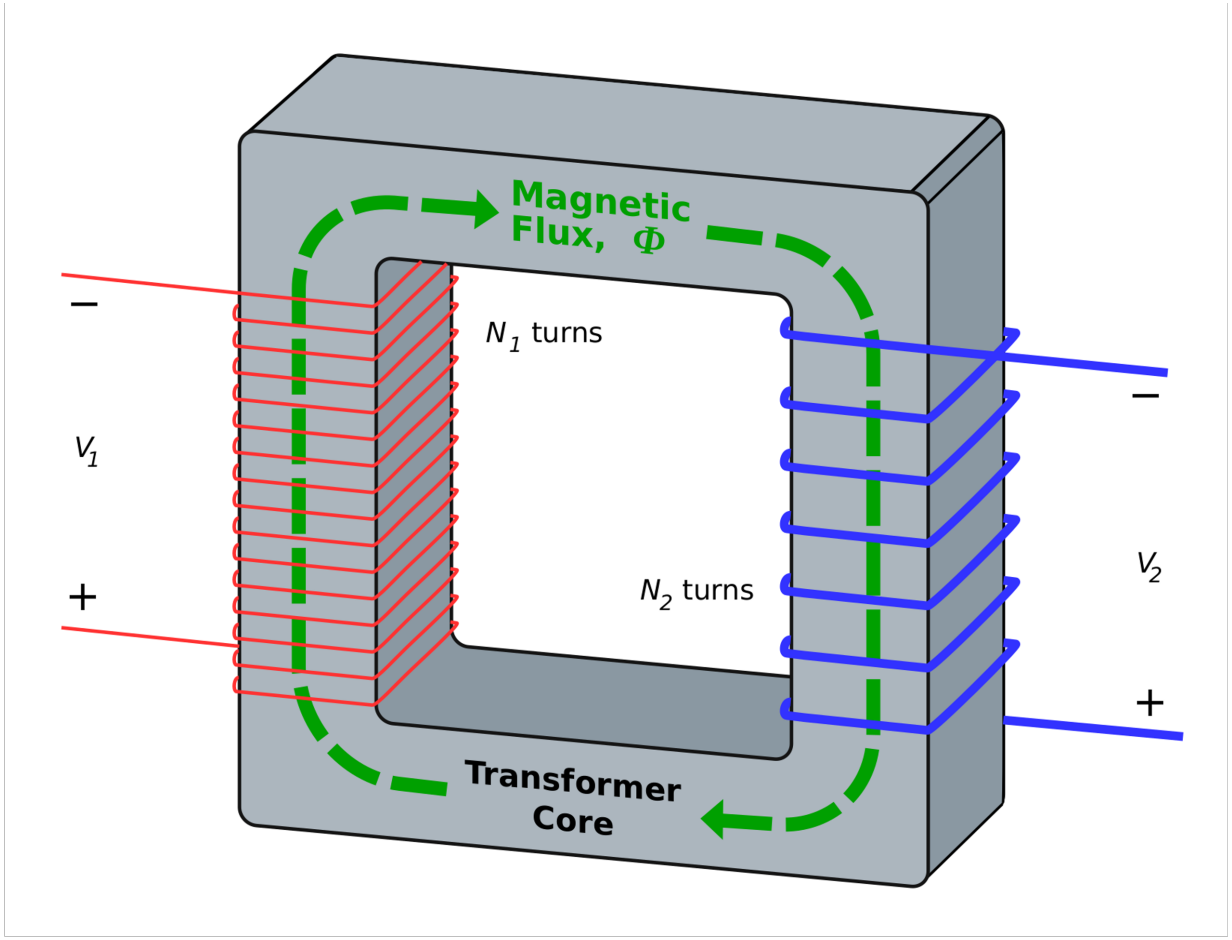


Fig. 8.8: © BillC, https://commons.wikimedia.org/wiki/File:Transformer3d col3.svg, CC BY SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0/). Modified by author.